

Maven Toys Sales & Inventory Operations Project

Description

This Confluence space documents the business analysis layer of the Maven Toys Sales & Inventory Operations Optimization project. It connects the business problem, stakeholder needs, requirements, KPIs, use cases, Jira user stories, SQL logic, Power BI dashboard pages, assumptions, limitations, and business recommendations into one structured BA portfolio case.

Search this Space

 Search

Filter by Label

 These are all the labels in use in this space. Select a label to see a list of content the label has been applied to.

Label list

As you and your team label content this area will fill up and display the latest updates.



Recently updated content

 This list below will automatically update each time somebody in your space creates or updates content.

-  [5.3 Business Recommendations](#)
3 minutes ago • contributed by [Bui Tuan Dung](#)
-  [5.1 Dashboard Evidence & Page Mapping](#)
about an hour ago • contributed by [Bui Tuan Dung](#)
-  [2.4 Requirement Traceability Matrix](#)

about an hour ago • contributed by [Bui Tuan Dung](#)

4.2 Jira Backlog Evidence

about an hour ago • contributed by [Bui Tuan Dung](#)

4.1 User Stories & Acceptance Criteria

about an hour ago • contributed by [Bui Tuan Dung](#)

3. Process & Use Cases

May 28, 2026 • contributed by [Bui Tuan Dung](#)

1.1 Project Home

May 28, 2026 • contributed by [Bui Tuan Dung](#)

3.2 Inventory Risk Review Decision Flow

May 28, 2026 • contributed by [Bui Tuan Dung](#)

1.3 Stakeholder Map

May 28, 2026 • contributed by [Bui Tuan Dung](#)

5. Dashboard, Insights & Recommendations

May 27, 2026 • contributed by [Bui Tuan Dung](#)

1. Project Foundation

 This space documents the business analysis layer of the Maven Toys Sales & Inventory Operations Optimization project. It connects the business problem, stakeholder needs, requirements, KPIs, use cases, Jira user stories, SQL logic, Power BI dashboard pages, assumptions, limitations, and business recommendations into one structured BA portfolio case.

Page	Description
1. Project Foundation	This section introduces the project context, business problem, stakeholders, and overall BA case structure.
2. Business Requirements & Solution Scope	This section defines the business requirements, KPI logic, data scope, and requirement-to-dashboard traceability.
3. Process & Use Cases	This section documents the operational use cases and decision flows supported by the dashboard.
4. Agile Delivery Backlog	This section documents the Agile delivery layer of the project, including Jira epics, user stories, acceptance criteria, and backlog evidence.
5. Dashboard, Insights & Recommendations	This section connects the final Power BI dashboard outputs with business insights, interpretation boundaries, and stakeholder recommendations.

1.1 Project Home

 This page summarizes the Maven Toys Sales & Inventory Operations Optimization project as a Business Analysis and Business Intelligence portfolio case. The project connects business requirements, SQL logic, Power BI dashboard design, Agile delivery artifacts, and stakeholder recommendations.

Project Summary

The Maven Toys Sales & Inventory Operations Optimization project analyzes retail sales, product profitability, store performance, and inventory risk using the Mexico Toy Sales dataset. The goal is to support management decision-making across revenue performance, product strategy, store efficiency, and stockout-risk monitoring.

The project uses SQL Server for backend data modeling and business logic, Power BI for dashboard visualization, Confluence for BA documentation, Jira for Agile backlog management, and Draw.io for stakeholder and process diagrams.

Business Problem

Retail decision-makers need a consolidated view of sales performance, product profitability, store/location efficiency, and inventory risk. Without a structured dashboard and clear KPI definitions, stakeholders may focus only on revenue while missing low-margin products, inefficient store patterns, or stockout-risk exposure.

Project Objective

Build a SQL-backed Power BI dashboard and supporting BA documentation that helps stakeholders:

- Monitor executive revenue, profit, margin, sales volume, and inventory-risk KPIs.
- Identify product categories and products that drive profit and margin.
- Compare store locations by both total contribution and per-store efficiency.
- Detect out-of-stock and high-stockout-risk product-store combinations.
- Translate dashboard insights into practical business recommendations.

Tools Used

Tool	Purpose
SQL Server	Data validation, SQL views, KPI logic, and business analysis queries
Power BI	Interactive dashboard design and visual storytelling
Confluence	Business requirements, KPI dictionary, use cases, assumptions, limitations, and recommendations
Jira	Epics, user stories, acceptance criteria, and Agile delivery tracking
Draw.io	Stakeholder map and inventory risk review decision flow
GitHub	Public portfolio packaging and project repository

Main Stakeholders

Stakeholder	Main need
Executive Manager	Monitor overall business performance, profitability, and operational risk.
Product / Merchandising Manager	Understand product/category profitability, margin performance, and product prioritization.
Store Operations Manager	Compare store and location performance by total contribution and per-store efficiency.
Inventory Planner	Prioritize stockout-risk items and replenishment review.
Store Managers	Use store-level insights for local operational action.
Procurement / Buying Team	Support replenishment planning and supplier-order coordination.

BI / Data Analyst	Maintain SQL logic, KPI definitions, data model accuracy, and dashboard reliability.
Customers / Parents	Expect product availability, value, and safe products.
Toy Suppliers / Manufacturers	Provide product supply, cost inputs, and replenishment support.
Product Safety / Compliance	Ensure toy safety, labeling, and regulatory considerations are recognized.

Dashboard Pages

Dashboard page	Purpose
Executive Summary	Monitor high-level revenue, profit, margin, units sold, inventory value, and urgent inventory items
Product Profitability Analysis	Compare product/category revenue, profit, margin, units sold, and profit per unit
Store & Location Performance	Compare total store contribution and average performance per store
Inventory Risk & Replenishment	Monitor stock on hand, inventory value, stockout risk, and urgent replenishment items

Final Deliverables

Deliverables	Location
SQL scripts	GitHub / 02_sql
Power BI dashboard	GitHub / 03_powerbi

BA documentation	Confluence / GitHub export / 04_ba_docs
Stakeholder and process diagrams	Draw.io / GitHub / 05_diagrams
Jira backlog evidence	Jira / GitHub screenshots
Final case study	GitHub README

1.2 Business Context

 This page explains the business context, stakeholder pain points, decision needs, and why the Maven Toys dashboard solution was created.

Business Context

Maven Toys operates across multiple store locations and sells products across several categories, including Toys, Electronics, Art & Crafts, Games, and Sports & Outdoors. Management needs to understand not only how much revenue the business generates, but also which products, stores, and inventory areas require attention.

A revenue-only view is not sufficient because high revenue does not always indicate strong profitability. Similarly, a high total inventory value does not necessarily mean that the right products are available in the right stores. The project therefore combines sales, profit, store, and inventory-risk analysis into one decision-support dashboard.

Business Pain Points

Pain Point	Business impact
Revenue alone can be misleading	High-revenue products may have weaker margins and lower profit efficiency
Store count can distort location performance	Locations with more stores may appear stronger by total revenue even if per-store efficiency is lower
Inventory risk is difficult to prioritize manually	Out-of-stock and high-risk product-store combinations may be missed
KPI definitions may be inconsistent	Stakeholders may interpret revenue, profit, margin, inventory value, and risk differently
Inventory data appears to be a current stock snapshot	Inventory-risk results require careful interpretation and documented assumptions

Key Business Questions

Area	Business question
Executive Performance	How is the business performing overall in revenue, profit, margin, sales volume, and inventory exposure?
Product Profitability	Which categories and products generate the most profit and strongest margin?
Product Strategy	Are high-revenue products also high-margin products?
Store Performance	Which store locations perform best by total contribution and per-store efficiency?
Inventory Risk	Which product-store combinations are exposed to stockout risk?
Replenishment Review	Which inventory items should be prioritized for operational follow-up?

Decision Needs

Decision output	Business purpose
Executive Performance View	Give management a high-level view of business health and operational risk
Product Action Matrix	Help product stakeholders classify products by revenue, profit, margin, and profit per unit

Store Efficiency Benchmark	Help operations stakeholders separate total location contribution from per-store efficiency
Inventory Risk Priority	Help inventory stakeholders prioritize out-of-stock and high-stockout-risk items

Why This Project Matters

This project matters because it converts raw sales and inventory data into a structured business review tool. Instead of only reporting what happened, the dashboard supports practical management decisions around product prioritization, store benchmarking, and replenishment review.

The BA documentation layer ensures that the dashboard is connected to stakeholder needs, business requirements, KPI definitions, use cases, assumptions, and Agile delivery artifacts.

1.3 Stakeholder Map

 This page contains the stakeholder map created in Draw.io and explains each stakeholder's role in the dashboard solution.

Stakeholder group	Stakeholders	Role in the Project	Business concern
Core Dashboard Users	Executive Manager	Primary business sponsor	Needs high-level visibility into revenue, profit, margin, sales volume, and operational risk.
Core Dashboard Users	Product / Merchandising Manager	Product and assortment decision stakeholder	Needs product/category profitability, margin performance, and product prioritization insights.
Core Dashboard Users	Store Operations Manager	Store and location performance stakeholder	Needs store efficiency, location benchmarking, and performance comparison.
Core Dashboard Users	Inventory Planner	Inventory-risk and replenishment-review stakeholder	Needs stockout-risk prioritization and inventory review categories.
Supporting Operational Stakeholders	Store Managers	Local execution stakeholder	Use store-level insights to support local stock, product, and operational actions.
Supporting Operational	Procurement / Buying Team	Supplier-order and buying stakeholder	Supports replenishment

Stakeholders			planning, supplier coordination, and product availability review.
Supporting Operational Stakeholders	BI / Data Analyst	Technical delivery and governance owner	Maintains SQL logic, KPI definitions, data model reliability, and dashboard accuracy.
External & Governance Stakeholders	Customers / Parents	External impacted stakeholder	Expect product availability, value, relevant assortment, and safe products.
External & Governance Stakeholders	Toy Suppliers / Manufacturers	External supply stakeholder	Provide product supply, cost inputs, and replenishment support.
External & Governance Stakeholders	Product Safety / Compliance	Governance and product-risk stakeholder	Ensures toy safety, labeling, and regulatory considerations are recognized.

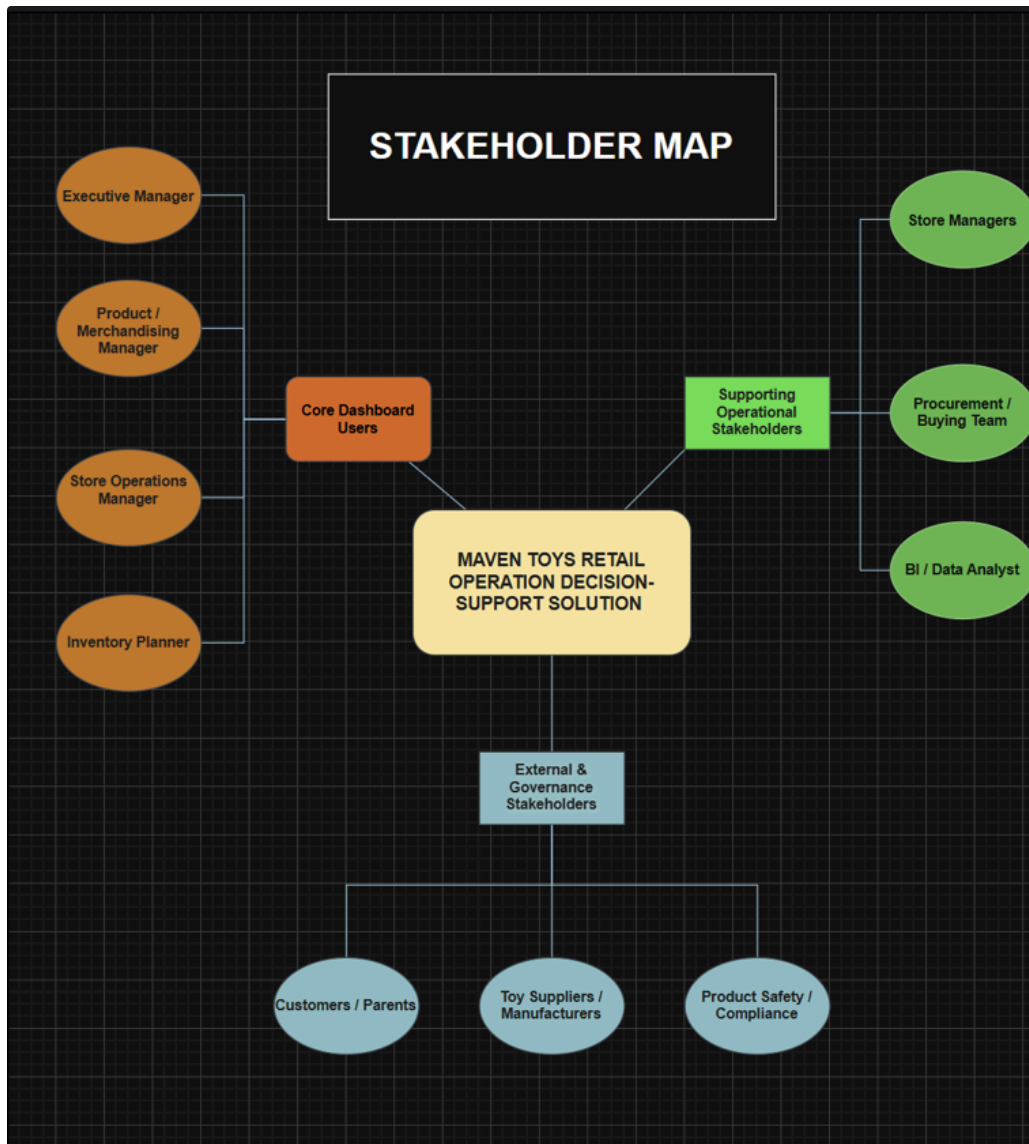


Figure 1. Stakeholder Map for the Maven Toys Retail Operations Decision-Support Solution

The stakeholder map reflects a realistic toy-retail operating environment. The Maven Toys Retail Operations Decision-Support Solution connects core dashboard users, supporting operational stakeholders, and external/governance stakeholders.

Core dashboard users include executive, product/merchandising, store operations, and inventory stakeholders who use the dashboard to monitor revenue, profitability, store performance, and stockout risk.

Supporting operational stakeholders include store managers, procurement/buying teams, and BI analysts who help translate dashboard outputs into local action, replenishment planning, and data governance.

External and governance stakeholders include customers/parents, toy suppliers/manufacturers,

and product safety/compliance considerations. Customers and parents are not direct dashboard users, but they are indirectly affected through product availability, assortment quality, and toy safety expectations.

2. Business Requirements & Solution Scope

 This section defines the business requirements, KPI logic, data scope, and requirement-to-dashboard traceability for the Maven Toys Sales & Inventory Operations project.

Sub-pages in this section

Sub-page	Purpose
2.1 Business Requirements Document	Documents the business problem, objectives, scope, functional requirements, non-functional requirements, and success criteria.
2.2 KPI Dictionary	Defines each dashboard KPI, including business meaning, formula, source view, dashboard page, and interpretation.
2.3 Data Dictionary	Documents project-relevant tables, views, fields, and business meanings used in SQL and Power BI.
2.4 Requirement Traceability Matrix	Connects business objectives, requirements, Jira epics, SQL objects, Power BI pages, and evidence.

2.1 Business Requirements Document (BRD)

 This page documents the business requirements, functional requirements, non-functional requirements, scope, and success criteria for the Maven Toys Sales & Inventory Operations Optimization project.

Background

Maven Toys needs a structured business intelligence solution to monitor sales performance, product profitability, store/location performance, and inventory risk. The dashboard is designed to support management review and operational prioritization rather than replace a full inventory optimization system.

Business Problem

Retail stakeholders may over-rely on revenue rankings when evaluating business performance. This can hide lower-margin products, distort store performance when store count differs by location, and overlook product-store combinations exposed to stockout risk. A requirement-driven dashboard is needed to connect KPIs, business questions, and operational actions.

Project Objectives

- Provide an executive overview of revenue, profit, margin, units sold, transactions, inventory value, and urgent inventory items.
- Identify product and category profitability patterns using revenue, profit, margin, units sold, and profit per unit.
- Compare store and location performance using both total contribution and average performance per store.
- Classify inventory risk and prioritize out-of-stock and high-stockout-risk product-store combinations.
- Document KPI definitions, assumptions, limitations, and business recommendations for stakeholder interpretation.

Scope

In Scope	Description
----------	-------------

Sales and profitability analysis	Revenue, cost, profit, profit margin, units sold, and transaction analysis
Product analysis	Category and product-level profitability, margin, and action matrix
Store analysis	Store and location performance by total and average metrics
Inventory-risk analysis	Inventory value, stock on hand, risk category, urgent items, and stockout risk rate
BA documentation	BRD, KPI dictionary, data dictionary, use cases, Jira user stories, assumptions, recommendations, and traceability

Out of Scope	Reason
Full replenishment optimization	Supplier lead time, reorder quantity, reorder point, and safety stock are not available
Demand forecasting	The current version focuses on descriptive and diagnostic decision support
Customer segmentation	The dataset does not contain customer-level basket or demographic data
Promotion effectiveness analysis	Promotion calendar and campaign fields are not available

Functional Requirements

ID	Requirement	Dashboard area
FR-01	The dashboard must show total revenue, total cost, total profit, profit margin, units sold, and transactions.	Executive Summary
FR-02	The dashboard must show monthly revenue and profit trends.	Executive Summary
FR-03	The dashboard must compare product categories and products by revenue, profit, margin, units sold, and profit per unit.	Product Profitability Analysis
FR-04	The dashboard must help users identify high-revenue but weaker-margin products.	Product Profitability Analysis
FR-05	The dashboard must compare store locations by total contribution and average performance per store.	Store & Location Performance
FR-06	The dashboard must classify inventory by risk category.	Inventory Risk & Replenishment
FR-07	The dashboard must highlight urgent inventory items, including out-of-stock	Inventory Risk & Replenishment

	and high-stockout-risk items.	
FR-08	The dashboard must allow filtering by year, store location, product category, and store city.	All pages

Non-Functional Requirements

ID	Requirement
NFR-01	KPI values must be validated against SQL query outputs.
NFR-02	Dashboard pages must use consistent page navigation, naming, and formatting.
NFR-03	KPI definitions must be documented in the KPI Dictionary.
NFR-04	Inventory-risk assumptions must be clearly documented because inventory data is treated as a current stock snapshot.
NFR-05	Dashboard visuals should support business decision-making, not only descriptive reporting.

Success Criteria

- Executive KPIs are visible and match SQL-validated values.
- Product, store, and inventory pages answer the defined business questions.
- Inventory-risk categories are clearly interpreted as action priorities.

- Stakeholders can trace dashboard pages back to requirements and Jira user stories.
- Assumptions and limitations are documented clearly enough to prevent overclaiming.

2.2 KPI Dictionary

 This page defines the KPIs used in the dashboard, including formulas, source views, dashboard pages, and business interpretation.

Cards name	Business definition	Formula / Logic	Source	Dashboard page	Business interpretation
Total Revenue	Total sales value	Units x Product Price	vw_sales_profit	Executive / Product / Store	Measures sales scale
Total Cost	Total product cost	Units x Product Cost	vw_sales_profit	Executive	Measures cost base
Total Profit	Revenue minus cost	Revenue - Cost	vw_sales_profit	All sales pages	Measures profitability
Profit Margin	Profit as percentage of revenue	Profit / Revenue	vw_sales_profit	Executive / Product / Store	Shows profitability efficiency
Units Sold	Total quantity sold	SUM Units	vw_sales_profit	Executive / Product / Store	Measures sales volume
Total Transactions	Number of sales rows	DISTINCTCOUNT Sale_ID	vw_sales_profit	Executive	Measures transaction volume
Profit per Unit	Profit generated per unit sold	Total Profit / Units Sold	Power BI measure	Product	Supports unit economics comparison
Average Revenue per Store	Revenue divided by store count	Revenue / distinct stores	Power BI measure	Store	Separates location scale from store efficiency

Average Profit per Store	Profit divided by store count	Profit / distinct stores	Power BI measure	Store	Shows per-store profitability efficiency
Inventory Value	Current stock value at cost	Stock On Hand x Product Cost	vw_inventory_risk	Inventory	Shows capital tied in inventory
Stock On Hand	Available inventory quantity	SUM Stock_On_Hand	vw_inventory_risk	Inventory	Measures current stock availability
Days of Inventory Left	Estimated stock coverage	Stock On Hand / Avg Daily Sales	vw_inventory_risk	Inventory	Supports replenishment review
High Stockout Risk Items	Product-store records at high stockout risk	COUNT rows where risk = High Stockout Risk	vw_inventory_risk	Inventory	Identifies short-term replenishment risk
Out of Stock Items	Product-store records with zero stock and historical demand	COUNT rows where risk = Out of Stock	vw_inventory_risk	Inventory	Identifies immediate availability issues
Urgent Inventory Items	High risk plus out-of-stock records	High Stockout Risk Items + Out of Stock Items	Power BI measure	Inventory	Prioritizes inventory action
Stockout Risk Rate	Share of product-store records that are urgent	Urgent Inventory Items / Product-Store Count	Power BI measure	Inventory	Measures overall stockout exposure

2.3 Data Dictionary

 This page documents the project-relevant tables, views, fields, and business meanings used in the SQL and Power BI model.

Field	Source	Meaning	Usage	Notes
Sale_ID	sales / vw_sales_profit	Sales transaction identifier	Transaction count	Used for total transactions
Date	sales / vw_sales_profit	Sales transaction date	Trend analysis	Connected to DimDate in Power BI
Product_ID	sales / products / inventory	Product key	Product and inventory analysis	Join key
Product_Name	products	Product display name	Product visuals	Used in product matrix and rankings
Product_Categor y	products	Product category grouping	Category analysis	Used in slicers and visuals
Store_ID	sales / stores / inventory	Store key	Store and inventory analysis	Join key
Store_Name	stores	Store display name	Store analysis	Used in store hierarchy
Store_City	stores	Store city	Store analysis	Used in filtering and hierarchy
Store_Location	stores	Store location type	Location analysis	Used for Airport, Downtown, and other location comparisons
Units	sales	Quantity sold	Revenue, cost, profit	Multiplied by price and cost

Product_Cost	products	Cost per unit	Cost, profit, inventory value	Used in SQL views
Product_Price	products	Selling price per unit	Revenue and profit	Used in SQL views
Revenue	vw_sales_profit	Calculated sales value	Sales KPIs	Units x Product Price
Cost	vw_sales_profit	Calculated product cost	Profit analysis	Units x Product Cost
Profit	vw_sales_profit	Revenue minus cost	Profitability analysis	Revenue - Cost
Stock_On_Hand	inventory / vw_inventory_risk	Current stock quantity	Inventory risk	Current stock snapshot
Inventory_Value	vw_inventory_risk	Stock value at cost	Inventory dashboard	Stock_On_Hand x Product_Cost
Avg_Daily_Sales	vw_inventory_risk	Estimated average product demand	Inventory risk	Used to calculate days left
Days_of_Inventory_Left	vw_inventory_risk	Estimated inventory coverage	Inventory risk	Directional stock coverage metric
Inventory_Risk	vw_inventory_risk	Risk category classification	Inventory dashboard	Rule-based classification

2.4 Requirement Traceability Matrix

 This page connects business objectives, functional requirements, Jira epics, Jira stories, SQL objects, Power BI dashboard pages, and evidence. It shows how each dashboard output is traceable back to a stakeholder need and business requirement.

Req. ID	Business objective	Requirement	Epic	Story	SQL	Power BI Page	Evidence
FR-01	Monitor executive business performance	Show total revenue, total profit, profit margin, units sold, inventory value, and urgent inventory items.	Executive Performance Overview	KAN-6 View executive KPI summary	vw_sales_profit; vw_inventory_risk	Executive Summary	KPI cards; Jira story KAN-6; KPI Dictionary
FR-02	Track sales and profit trend over time	Show monthly revenue and profit trend with correct Month-Year sorting.	Executive Performance Overview	KAN-7 Monitor monthly revenue and profit trend	vw_sales_profit; dimDate	Executive Summary	Monthly trend visual; Month-Year sorting evidence
FR-03	Support product/category profitability decisions	Compare product categories by revenue, profit, margin, units sold, and profit per unit.	Product Profitability Analysis	KAN-8 Compare product category profitability	vw_sales_profit; products	Product Profitability Analysis	Profitability matrix; category comparison visuals

FR-04	Identify product action opportunities	Identify high-revenue/high-margin, high-revenue/lower-margin, and lower-performing products.	Product Profitability Analysis	KAN-9 Identify product action groups	vw_sales_profit; products	Product Profitability Analysis	Revenue vs margin scatter; top product visual
FR-05	Compare store/location performance	Compare store locations by total contribution and per-store efficiency.	Store & Location Performance Analysis	KAN-10 Compare store locations by total contribution; KAN-11 Compare store locations by per-store efficiency	vw_sales_profit; stores	Store & Location Performance	Store hierarchy matrix; average revenue/profit per store visuals
FR-06	Classify inventory risk	Classify inventory into Out of Stock, High Stockout Risk, Medium Stockout Risk, Healthy Stock, and No Sales History.	Inventory Risk Review	KAN-12 Classify inventory risk	vw_inventory_risk	Inventory Risk & Replenishment	Inventory risk distribution; risk classification logic

FR-07	Prioritize urgent inventory review	Highlight out-of-stock and high-stockout-risk product-store combinations .	Inventory Risk Review	KAN-13 Prioritize urgent inventory review items	vw_inventory_risk	Inventory Risk & Replenishment	Urgent Inventory Items KPI; inventory risk by category/location
FR-08	Support responsible interpretation	Document assumptions, limitations, and data-quality boundaries.	Data Quality & KPI Governance	KAN-15 Validate sales-profit view; KAN-16 Document assumptions and limitations	vw_sales_profit; vw_inventory_risk; SQL validation scripts	All pages	Assumptions & Limitations page; SQL validation evidence

The traceability matrix confirms that each major dashboard component is linked to a documented business requirement, Jira delivery item, SQL/data object, and Power BI dashboard page. This helps ensure that the dashboard remains requirement-driven rather than visual-only, and that stakeholder needs can be traced through analysis, implementation, and evidence.

3. Process & Use Cases

 This section documents the operational use cases and decision flows supported by the dashboard.

Sub-pages in this section

Sub-page	Purpose
3.1 Use Cases	Documents the main business use cases, including executive monitoring, product analysis, store comparison, inventory risk identification, and replenishment review.
3.2 Inventory Risk Review Decision Flow	Contains the Draw.io decision flow showing how inventory-risk signals are converted into replenishment review actions.

3.1 Use Cases

 This page documents the main business use cases supported by the dashboard, including executive monitoring, product analysis, store comparison, and inventory risk review.

Use Case ID	Use Case	Primary Actor	Business Goal
UC-01	Monitor Executive Performance	Executive Manager	Understand overall revenue, profit, margin, sales volume, and inventory risk
UC-02	Analyze Product Profitability	Product Manager	Identify strategic winners, margin-review products, and lower-priority products
UC-03	Compare Store & Location Performance	Operations Manager	Separate total contribution from per-store efficiency
UC-04	Identify Inventory Stockout Risk	Inventory Planner	Find out-of-stock and high-risk product-store combinations
UC-05	Prioritize Weekly Replenishment Review	Inventory Planner	Convert risk signals into operational review actions

UC-01: Monitor Executive Performance

Field	Description
Primary Actor	Executive Manager
Goal	Understand overall business health and operational risk.

Preconditions	Sales-profit and inventory-risk data are refreshed and dashboard filters are available.
Main Flow	1. User opens Executive Summary page. 2. User reviews revenue, profit, margin, units sold, inventory value, and urgent inventory items. 3. User checks monthly revenue and profit trend. 4. User reviews risk distribution and high-level business signals.
Alternative Flow	If a KPI appears inconsistent, user refers to KPI Dictionary and SQL validation notes.
Postcondition	Executive user identifies the current business performance position and risk areas.
Business Value	Supports high-level management review and prioritization.

UC-02: Analyze Product Profitability

Field	Description
Primary Actor	Product Manager
Goal	Identify products and categories that drive profit and margin.
Preconditions	Product and sales-profit data are available in the dashboard.
Main Flow	1. User opens Product Profitability Analysis page. 2. User reviews product/category matrix. 3. User compares revenue, profit, margin, units sold, and profit per unit.

	4. User identifies strategic winners and margin-review products.
Alternative Flow	If a product has high revenue but low margin, user flags it for pricing, cost, or promotion review.
Postcondition	Product stakeholders identify product action groups.
Business Value	Supports product prioritization and margin management.

UC-03: Compare Store & Location Performance

Field	Description
Primary Actor	Operations Manager
Goal	Compare store performance by total contribution and per-store efficiency.
Preconditions	Store and sales-profit data are available in the dashboard.
Main Flow	1. User opens Store & Location Performance page. 2. User reviews total revenue and total profit by location. 3. User compares average revenue and average profit per store. 4. User checks store count context.
Alternative Flow	If a location leads in total performance but not average performance, user separates scale effect from efficiency.
Postcondition	Operations stakeholders identify scale contributors and efficiency benchmarks.

Business Value	Supports store performance review and location benchmarking.
----------------	--

UC-04: Identify Inventory Stockout Risk

Field	Description
Primary Actor	Inventory Planner
Goal	Identify product-store combinations that require replenishment attention.
Preconditions	Inventory-risk view has been refreshed and dashboard filters are available.
Main Flow	1. User opens Inventory Risk & Replenishment page. 2. User reviews inventory value, stock on hand, high stockout risk items, out-of-stock items, and stockout risk rate. 3. User filters by product category, store location, or store city. 4. User identifies urgent inventory items.
Alternative Flow	If an item has no sales history, user classifies it as a separate investigation case.
Postcondition	Urgent product-store inventory items are identified.
Business Value	Supports proactive replenishment review and reduces stockout exposure.

UC-05: Prioritize Weekly Replenishment Review

Field	Description
Primary Actor	Inventory Planner

Goal	Convert inventory-risk signals into an operational review list.
Preconditions	Inventory-risk categories and urgent item measures are available.
Main Flow	1. User filters inventory dashboard for Out of Stock and High Stockout Risk. 2. User reviews category and location distribution. 3. User exports or records priority items. 4. User prepares a weekly replenishment discussion list.
Alternative Flow	Medium risk items are added to a watchlist instead of immediate review.
Postcondition	A weekly urgent inventory review list is prepared.
Business Value	Converts dashboard insight into operational action.

3.2 Inventory Risk Review Decision Flow

i This page contains the Draw.io decision flow showing how inventory risk signals are approached and reviewed.

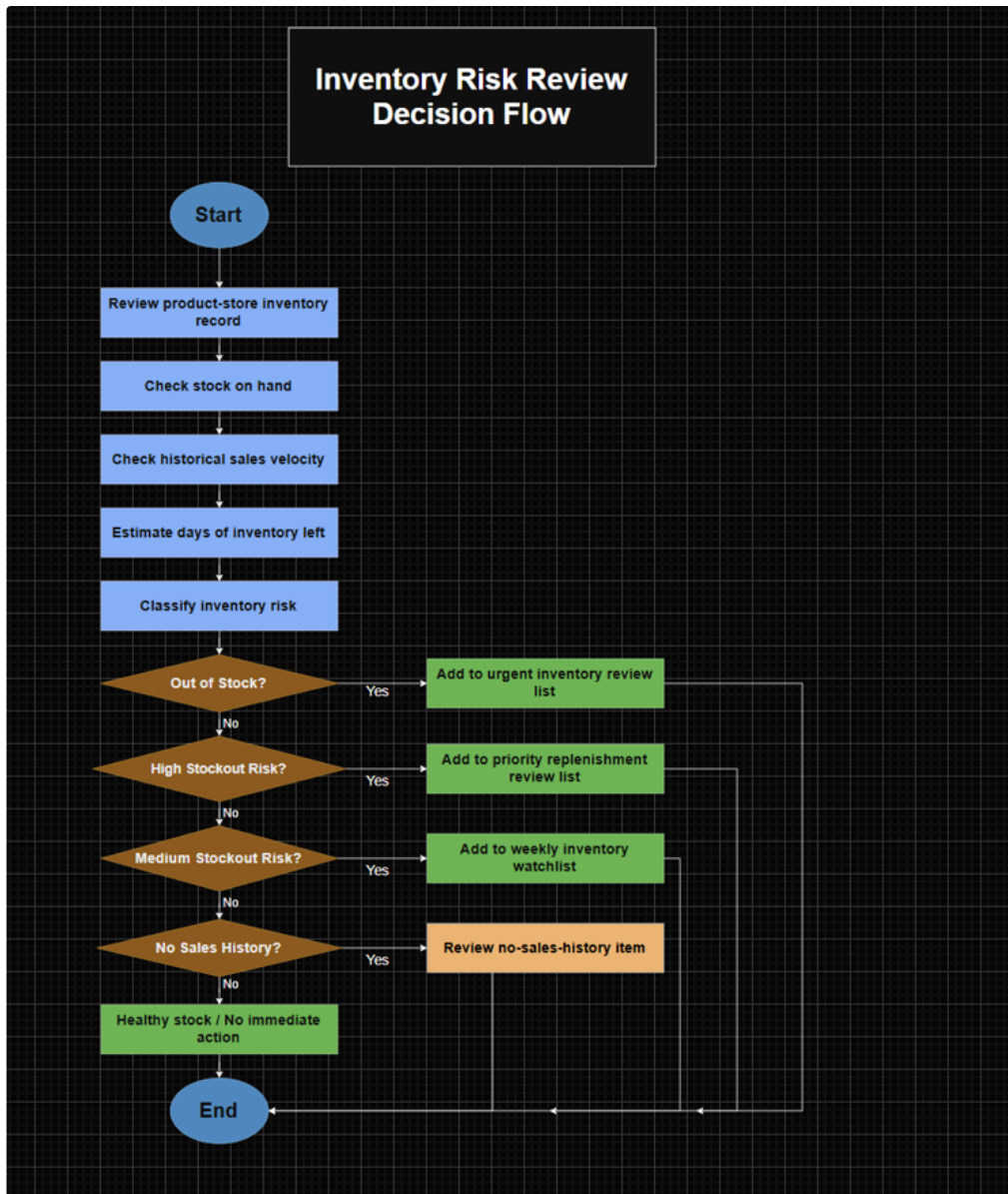


Figure 2. Inventory Risk Review Decision Flow

The inventory risk review decision flow translates dashboard risk signals into operational review categories. Product-store inventory records are reviewed using stock on hand, historical sales velocity, and estimated days of inventory left.

Out-of-stock items are added to an urgent inventory review list, high-risk items are added to a priority replenishment review list, medium-risk items are added to a weekly inventory watchlist, no-sales-history items are reviewed separately, and healthy stock requires no immediate action.

This flow is intentionally designed as a decision-support process rather than an automatic replenishment process. Final replenishment decisions still require business review, supplier information, lead time, reorder quantity, and operational judgment.

Terms interpretation

Risk condition	Action category	Interpretation
Out of Stock	Add to urgent inventory review list	Requires immediate review, but not always automatic reorder. The team may need to validate inventory records, check supplier availability, transfer stock, or review whether the product is still active.
High Stockout Risk	Add to priority replenishment review list	Likely requires replenishment or stock transfer review soon because estimated stock coverage is low.
Medium Stockout Risk	Add to weekly inventory watchlist	Should be monitored before it becomes urgent.
No Sales History	Review no-sales-history item	Demand cannot be estimated reliably; review whether the item is newly stocked, slow-moving,

		mismatched to the store, or affected by data coverage issues.
Healthy Stock	No immediate action	Stock level appears acceptable based on current assumptions and available demand history.

4. Agile Delivery Backlog

 This section documents the Agile delivery layer of the project, including Jira epics, user stories, acceptance criteria, and backlog evidence.

Sub-pages in this section

Sub-page	Purpose
4.1 User Stories & Acceptance Criteria	Summarizes Jira epics, user stories, and acceptance criteria that translate business requirements into Agile delivery items.
4.2 Jira Backlog Evidence	Stores screenshots and notes from the Jira board, including epics, backlog items, user stories, and acceptance criteria examples.

4.1 User Stories & Acceptance Criteria

 Jira Kanban was used to track this individual portfolio project. The backlog is structured with Agile/Scrum-compatible epics, user stories, acceptance criteria, priorities, and subtasks.

Epic	Story	Stakeholder	Priority	Acceptance criteria
Executive Performance Overview	KAN-6 View executive KPI summary	Executive Manager	Highest	Dashboard shows executive KPIs; KPI values match SQL outputs; users can filter by year, location, category, and city; KPI definitions are documented.
Executive Performance Overview	KAN-7 Monitor monthly revenue and profit trend	Executive Manager	Medium	Dashboard includes monthly revenue/profit trend; month-year sorting is correct; values align with SQL and Power BI measures.
Product Profitability Analysis	KAN-8 Compare product category profitability	Product / Merchandising Manager	High	Dashboard shows category-level revenue, profit, margin, units sold, and profit per unit; users can rank categories; KPI logic is documented.
Product Profitability Analysis	KAN-9 Identify product action groups	Product / Merchandising Manager	High	Dashboard supports interpretation of high-revenue/high-

				margin, high-revenue/lower-margin, and low-performing products; recommendations are documented.
Store & Location Performance Analysis	KAN-10 Compare store locations by total contribution	Store Operations Manager	Medium	Dashboard shows total revenue and profit by store location; store hierarchy supports comparison; results align with validated measures.
Store & Location Performance Analysis	KAN-11 Compare store locations by per-store efficiency	Store Operations Manager	High	Dashboard shows average revenue/profit per store; store count is visible; Downtown versus Airport interpretation is documented.
Inventory Risk Review	KAN-12 Classify inventory risk	Inventory Planner	Highest	Inventory is classified into risk groups; logic uses stock on hand, sales velocity, and days left; definitions and assumptions are documented.
Inventory Risk Review	KAN-13 Prioritize urgent inventory review items	Inventory Planner	Highest	Dashboard highlights Out of Stock and High Stockout Risk items;

				urgent items can be filtered; supports weekly inventory review.
Inventory Risk Review	KAN-14 Review no-sales-history inventory items	Inventory Planner	Medium	Dashboard separates No Sales History items; they are not treated as automatic stockout risk; interpretation is documented in the decision flow.
Data Quality & KPI Governance	KAN-15 Validate sales-profit view	BI / Data Analyst	Highest	vw_sales_profit preserves all sales rows; revenue, cost, and profit formulas are documented; KPI values are validated against SQL outputs.
Data Quality & KPI Governance	KAN-16 Document assumptions and limitations	Project Stakeholders	High	Inventory snapshot assumption and limitations are documented; inventory-risk logic is positioned as decision support rather than full optimization.

Priority Rationale

Priority	Justifications	Stories
----------	----------------	---------

Highest	Critical foundation or core decision feature. If missing, the dashboard loses major business value or KPI trust.	KAN-6, KAN-12, KAN-13, KAN-15
High	Strong business value and important stakeholder decision support, but not the absolute foundation.	KAN-8, KAN-9, KAN-11, KAN-16
Medium	Useful supporting analysis or interpretation, but less critical than core KPI/risk logic.	KAN-7, KAN-10, KAN-14

Selected Subtask Breakdown

Subtasks were added only to selected high-value or technically complex stories where the work required multiple delivery steps across SQL validation, Power BI development, documentation, or evidence capture. Simpler stories were kept at the user-story level to avoid overcomplicating the Jira backlog.

Story	Reason for Subtasks
KAN-6 View executive KPI summary	Requires KPI validation, dashboard card creation, slicer checking, documentation, and screenshot evidence.
KAN-11 Compare store locations by per-store efficiency	Requires validation of average revenue/profit per store, store-count context, business interpretation, and screenshot evidence.
KAN-12 Classify inventory risk	Requires risk-category definition, SQL validation, Power BI visuals, KPI documentation, and decision-flow evidence.
KAN-15 Validate sales-profit view	Requires row-count validation, revenue/cost/profit formula checks, and documentation of KPI reliability.

4.2 Jira Backlog Evidence

 This page provides Jira evidence showing how the Maven Toys project requirements were translated into epics, user stories, acceptance criteria, priorities, selected subtasks, and Kanban workflow statuses.

Evidence Checklist

Items	Screenshot File	Caption	Proof
Figure 1	jira_board_status_overview.png	Jira Kanban Board Status Overview	Shows workflow status distribution across To Do, In Progress, In Review, and Done.
Figure 2	jira_list_epics_stories_priorities.png	Jira List View with Epics, Stories, Priorities, and Statuses	Shows Epic → Story structure, priorities, statuses, parent mapping, and assignee fields.
Figure 3	jira_epic_child_stories_inventory_risk.png	Inventory Risk Review Epic with Child Stories	Shows the parent-child hierarchy under the Inventory Risk Review Epic.
Figure 4	jira_story_acceptance_criteria_inventory_risk.png	Inventory Risk Story with Acceptance Criteria	Shows a user story written in Agile BA format with testable acceptance criteria.
Figure 5	jira_story_subtasks_example.png	Selected Story Subtasks Example	Shows delivery breakdown for a selected high-value or technically complex story.

Evidence Placement

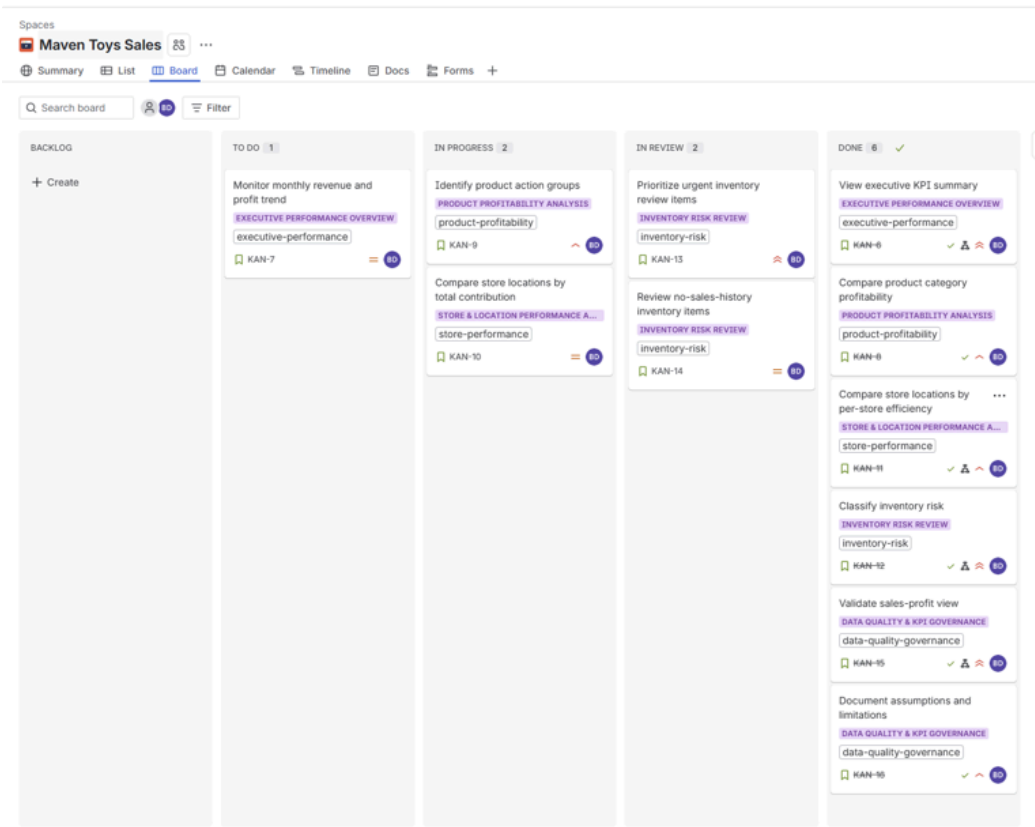


Figure 1. Jira Kanban Board Status Overview

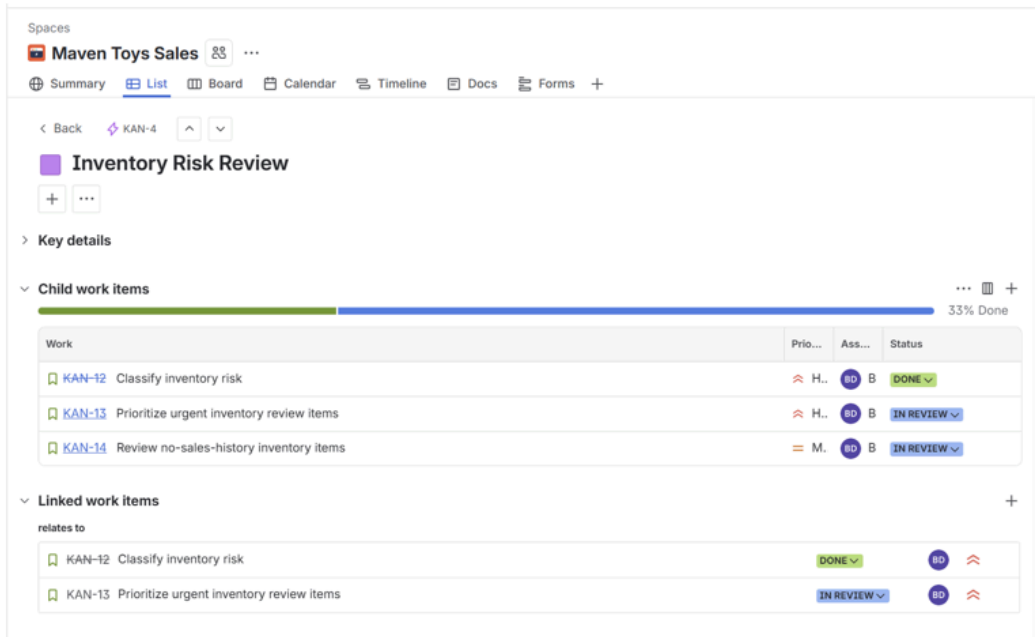


Figure 2. Jira List View with Epics, Stories, Priorities, and Statuses

Classify inventory risk



Key details

Environment

Add text

Original estimate

0m

Priority

🔼 Highest

Description

User Story:

As an Inventory Planner, I want inventory records classified into risk categories, so that I can prioritize inventory review actions.

Acceptance Criteria:

- Inventory is classified into Out of Stock, High Stockout Risk, Medium Stockout Risk, Healthy Stock, and No Sales History.
- Risk logic uses stock on hand, historical sales velocity, and estimated days of inventory left.
- Risk-category definitions are documented in the KPI Dictionary.
- Assumptions and limitations are documented in Confluence.

Subtasks



0% Done

Work	Priority	Assignee	Status
KAN-27 Define inventory risk categories Validate inventory-risk SQL logic	🔼 High	Unassigned	BACKLOG
KAN-28 Validate inventory-risk SQL logic	🔼 High	Unassigned	BACKLOG
KAN-29 Build inventory risk visuals in Power BI	🔼 High	Unassigned	BACKLOG
KAN-30 Document risk logic in KPI Dictionary	🔼 High	Unassigned	BACKLOG
KAN-31 Create inventory risk review decision flow	🔼 High	Unassigned	BACKLOG

Figure 5. Selected Story Subtasks Example

5. Dashboard, Insights & Recommendations

 This section presents the dashboard outputs, assumptions, limitations, and business recommendations.

Sub-pages in this section

Sub-page	Purpose
5.1 Dashboard Evidence & Page Mapping	Maps each Power BI dashboard page to stakeholder needs, business questions, related requirements, and Jira epics.
5.2 Assumptions & Limitations	Documents project assumptions, data limitations, and interpretation boundaries for the dashboard and inventory-risk model.
5.3 Business Recommendations	Translates dashboard findings into business recommendations for product strategy, store performance, and inventory-risk prioritization.

5.1 Dashboard Evidence & Page Mapping

 This page maps each Power BI dashboard page to the stakeholder need, business question, related requirement, and Jira epic it supports. It provides evidence that the dashboard is not only a visual report, but a requirement-driven decision-support solution.

Dashboard page	Primary stakeholder	Business question	Related requirements	Related Epic	Evidence / Page Notes
Executive Summary	Executive Manager; Inventory Planner	How is the business performing overall, and where is operational risk visible?	FR-01; FR-02; FR-07	Executive Performance Overview; Inventory Risk Review	KPI cards for revenue, profit, margin, units sold, inventory value, and urgent inventory items; revenue/profit trend; inventory risk distribution.
Product Profitability Analysis	Product / Merchandising Manager	Which categories and products drive profit, margin, units sold, and profit per unit?	FR-03; FR-04	Product Profitability Analysis	Revenue vs margin scatter; category-to-product profitability matrix; top products by profit; category contribution to monthly profit.

Store & Location Performance	Store Operations Manager; Store Managers	Which store locations perform best by total contribution and by per-store efficiency?	FR-05	Store & Location Performance Analysis	Store hierarchy matrix; average revenue/profit per store; store count mix; revenue vs margin positioning.
Inventory Risk & Replenishment	Inventory Planner; Procurement / Buying Team	Which product-store combinations require urgent inventory review or weekly monitoring ?	FR-06; FR-07; FR-08	Inventory Risk Review; Data Quality & KPI Governance	Inventory value by category; risk by store location; risk by product category; urgent inventory items; no-sales-history interpretation.

Data Visualization Placement

The dashboard evidence confirms that each Power BI page is connected to stakeholder decision needs and documented requirements. This page should be used together with the Requirement Traceability Matrix to show requirement-to-output alignment.

Figure Caption	Records	Purpose
Figure 1. Executive Summary Dashboard Page		Shows executive KPI summary, trend, profit drivers, and high-level inventory risk distribution.

Figure 2. Product Profitability Analysis Dashboard Page



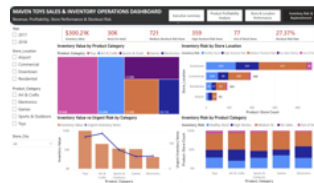
Shows product/category profitability, margin positioning, and top product contribution.

Figure 3. Store & Location Performance Dashboard Page



Shows total location contribution and per-store efficiency comparison.

Figure 4. Inventory Risk & Replenishment Dashboard Page



Shows stockout risk classification, urgent inventory items, inventory value, and risk distribution.

5.2 Assumptions & Limitations

 This page documents the assumptions, data limitations, and interpretation boundaries of the Maven Toys Sales & Inventory Operations Optimization project. It clarifies how the dashboard should be used and where additional operational data would be required for stronger replenishment decisions.

Assumptions

Assumption	Interpretation
Inventory data is treated as a current stock snapshot.	Inventory-risk results reflect current stock availability rather than historical stock movement.
Sales history is used as demand evidence.	Historical units sold are used to estimate sales velocity and days of inventory left.
Product price and product cost are treated as valid for KPI calculation.	Revenue, cost, profit, and margin are calculated from the available product master fields.
Inventory risk is directional.	Risk categories help prioritize review but do not replace a full replenishment planning system.

Limitations

Limitation	Business impact
No supplier lead time	Cannot calculate precise reorder timing.
No reorder point	Cannot define formal stock trigger thresholds.
No reorder quantity	Cannot recommend exact replenishment quantity.
No safety stock	Cannot account for demand uncertainty or buffer stock.

No daily historical inventory snapshots	Cannot analyze stock movement over time.
No lost sales data	Cannot quantify revenue lost due to stockouts.
No promotion calendar	Cannot separate normal demand from promotion-driven demand.
No product launch dates	Newer products may have shorter sales histories and require careful interpretation.
Some sales product-store combinations are missing current inventory records	Sales-profit analysis can retain these rows, but inventory-risk analysis should be interpreted cautiously.

Interpretation boundary

The inventory-risk model should be treated as a decision-support layer. It helps stakeholders identify likely stockout exposure and review priorities, but final replenishment decisions should incorporate supplier lead time, reorder rules, safety stock, promotion plans, and operational constraints.

5.3 Business Recommendations

 This page translates dashboard findings into business recommendations for product strategy, store performance management, and inventory-risk prioritization. Recommendations are linked to dashboard evidence and intended stakeholder actions.

Recommendation	Findings	Stakeholder	Priority
Establish a weekly urgent inventory review process for product-store records flagged as Out of Stock or High Stockout Risk.	The dashboard identifies 436 urgent inventory items: 359 High Stockout Risk items and 77 Out of Stock items. Stockout Risk Rate is 27.37%.	Inventory Planner; Store Operations Manager	Highest
Treat inventory-risk outputs as review priorities, not automatic replenishment orders.	Inventory value is about \$300.21K, but the dataset lacks supplier lead time, reorder quantity, safety stock, lost sales, promotion calendar, and daily inventory history.	Inventory Planner; BI / Data Analyst	Highest
Use Airport stores as the efficiency benchmark instead of judging locations by total contribution only.	Downtown has the largest total profit partly because it has 29 stores. Airport has the highest average profit per store (\$126,016.33) and highest margin (29.31%).	Store Operations Manager	High
Separate total location contribution from per-	Downtown leads by total revenue/profit,	Executive Manager; Store Operations	High

store efficiency when making store benchmarking or expansion decisions.	while Airport has fewer stores but stronger average revenue, average profit, and margin performance.	Manager	
Review high-revenue but lower-margin products separately and protect high-margin profit drivers.	Lego Bricks generates high revenue but has a low margin of 12.50%. Colorbuds is the strongest product by profit (\$834,944) and margin (53.37%).	Product / Merchandising Manager	High
Monitor the Toys category as both a profit driver and an inventory exposure area.	Toys is the highest-profit category and also has the highest inventory value at about \$99.86K.	Product / Merchandising Manager; Inventory Planner	High

Final business interpretation

The project shows that strong retail decision-making requires more than revenue tracking. Stakeholders should evaluate products through revenue, profit, margin, and unit economics; compare stores using both total contribution and per-store efficiency; and monitor inventory risk at product-store level instead of relying only on total inventory value.